Now, before we embark on the practical examples, I want to clarify a few facts and answer a few questions about embeddings.

WHAT DO EMBEDDINGS LOOK LIKE?

Let's explore some examples to get a sense of how embeddings are visually represented:

- *Word embeddings*—These are used to represent words as vectors in a low-dimensional space. These embeddings capture relationships between words. When visualized in 2D or 3D space, related words tend to cluster together.
  For example, a word embedding might look something like this:

```
const catEmbedding = [0.23, -0.56, 0.12, 0.87, -0.34, ...]
```

  While these numbers appear random to humans, they encode rich semantic information. Figure 5.9 is taken from tensorflow.org (https://projector.tensorflow.org/), demonstrating a more realistic example:

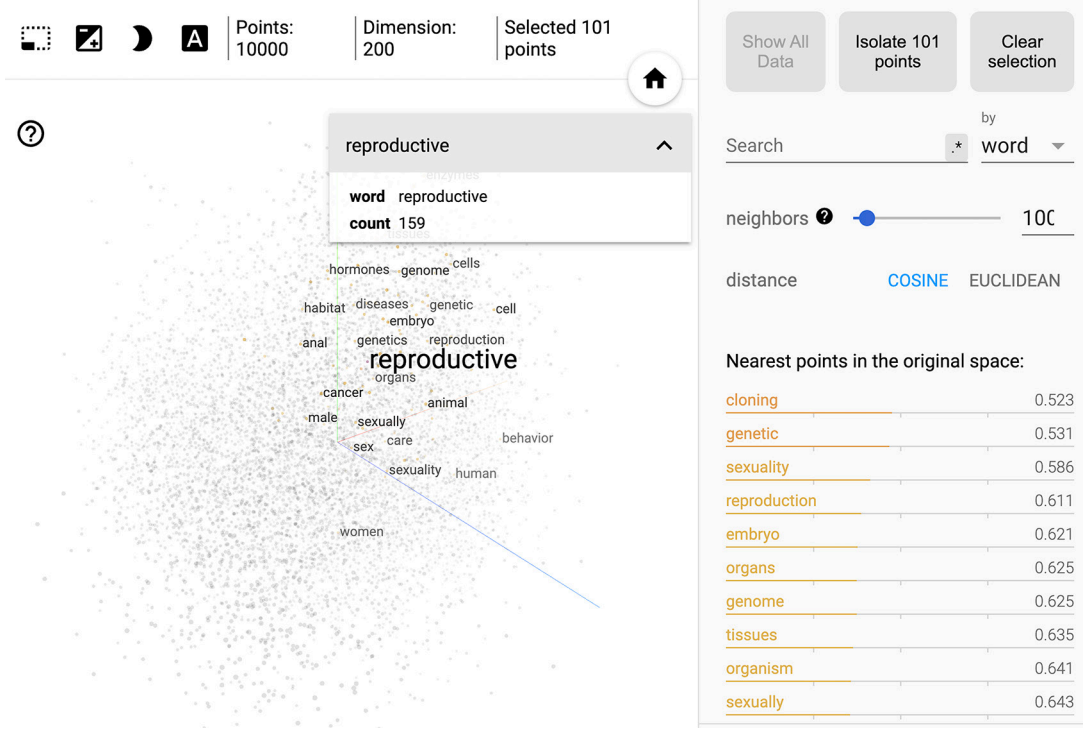


Figure 5.9 A 2D projection of word embeddings in a 200-dimensional space, focusing on the term "reproductive" and its semantic neighbors. The visualization shows 10,000 points, with 101 points selected for detailed view. The right panel lists the 10 nearest neighbors to "reproductive" in the original space, with their cosine distances. This representation demonstrates how word embeddings capture semantic relationships together in the vector space. (https://projector.tensorflow.org/)